

Fixed Point Theorems

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Contents

1	Introduction	2
2	Sperner's Lemma	2
2.1	Notation	2
2.2	Proof of Sperner's Lemma	3
3	Brouwer's Theorem	4
3.1	Proof of Brouwer's Theorem	4
3.2	The Other Direction	6
3.3	An Elementary Analytic Proof	6
4	Borsuk–Ulam Theorem and Equivalences	8
4.1	Borsuk–Ulam Theorem	8
4.2	Equivalences	8
5	Applications of Fixed Point Theorems	9
5.1	Necklace Splitting Problem	9
5.2	Ham Sandwich Theorem	10
5.3	Existence of Nash Equilibria	10
5.4	Cake and Rent Division	13
6	Computing Nash Equilibria	14
6.1	Complexity Analysis	14
6.2	Scarf's Algorithm	15

1 Introduction

A fixed point of a function $f : X \rightarrow X$ is a point $x_0 \in X$ such that $f(x_0) = x_0$. Fixed point theory is the field of mathematics concerned with the existence of fixed points. Brouwer's Fixed Point Theorem and the Borsuk–Ulam Theorem are well known for the way they relate fixed points to topology. Typical proofs of both theorems involve analysis and homology. In this paper, we first provide an elementary proof of Brouwer's Fixed Point Theorem, before demonstrating a sophisticated combinatorial proof using Sperner's Lemma, and give an overview of the equivalences between the topological and combinatorial analogues. Using the constructive combinatorial proofs provided, we also give several applications of these theorems, both to toy problems and real-world economic applications.

2 Sperner's Lemma

2.1 Notation

Definition 1. An n -simplex is an n -dimensional polytope that is the convex hull of its $n + 1$ vertices. Specifically, consider $n + 1$ points $v_1, \dots, v_{n+1} \in \mathbb{R}^N$, where $N \geq n$. These points are affinely independent, meaning the vectors $v_2 - v_1, v_3 - v_1, \dots, v_{n+1} - v_1$ are linearly independent. Then the n -simplex is the set of points

$$\Delta^n = \left\{ \sum_{i=1}^{n+1} \alpha_i v_i \mid \alpha_i \geq 0, \sum_{i=1}^{n+1} \alpha_i = 1 \right\}.$$

Definition 2. A k -face of Δ^n is a k -simplex formed by $k + 1$ vertices of Δ^n .

Example 3. A 0-face is a point. A 1-face is a line segment. A 2-face is a triangle. A 3-face is a tetrahedron, and so on.

Definition 4. A *simplicial subdivision* of Δ^n is a division of Δ^n into smaller n -dimensional simplices, called *cells*, such that the union of all cells covers Δ^n , and any two cells either intersect at a shared k -face for some $k < n$ or do not intersect at all.

Remark 5. An intuitive way to think about simplicial subdivisions is as a generalization of triangulations to higher dimensions.

Definition 6. A *Sperner coloring* of a simplicial subdivision of Δ^n is a labeling of the vertices of the subdivision using $n + 1$ colors such that:

1. no two of the $n + 1$ vertices of Δ^n share a color;
2. if a vertex of the subdivision lies on a k -face of Δ^n , it must be assigned one of the colors of the $k + 1$ vertices forming that face.

Our combinatorial proof also uses elementary graph theory terminology and lemmas.

Definition 7. Given a graph G with vertex set V and a vertex $v \in V$, the *degree* of v , denoted $\deg(v)$, is the number of edges incident to that vertex.

Definition 8. The *cell graph* of a simplicial subdivision of Δ^n is the graph with vertex set V , where each cell is assigned a vertex v , and the outer area of $\mathbb{R}^N$ is assigned the vertex v_{out} .

Definition 9. A *walk* on a graph is a sequence of vertices where each consecutive vertex is connected by an edge. A *trail* is a walk that does not visit the same edge twice.

2.2 Proof of Sperner's Lemma

Let $[k]$ denote the set of integers $\{1, 2, \dots, k\}$, V be the set of vertices in the simplicial subdivision of Δ^n , and $c : V \rightarrow [n + 1]$ be a valid Sperner coloring. For any subset $S \subseteq V$, we define the labeling function $L(S)$ as the set of all colors assigned to the vertices in S :

$$L(S) = \{c(v) \mid v \in S\}.$$

Definition 10. A k -face F of a coloring of the vertices is *rainbow* if and only if $L(F) = [k + 1]$.

Lemma 11 (Sperner's Lemma [Spe28]). Every Sperner coloring of a simplicial subdivision of Δ^n contains an odd number of *rainbow* cells.

Proof. We use a constructive argument on the trails around the vertices of the cell graph. Although the necessary base case $n = 1$ is best explained through algebraic methods, we will provide the intuition for the case $n = 2$, and then use induction to complete the proof for higher dimensions.

$n = 1$

The one-dimensional case of Sperner's Lemma is equivalent to saying that over consecutive line segments $[p_1, p_2], [p_2, p_3], \dots, [p_{m-1}, p_m]$, where $L(p_1) = 1$ and $L(p_m) = 2$, an odd number of line segments $[p_i, p_{i+1}]$ are rainbow. The following scenario is equivalent: keep a running value x starting at $x = 1$, define the function $f(x) = 3 - x$, and apply f to x each time we encounter a rainbow segment. Note $f^2(x) = x$, so if there were an even number of rainbow segments, we would end with $x = 1 \neq 2$, which is a contradiction.

$n = 2$

The two-dimensional case of Sperner's Lemma concerns a triangulation T of a triangle ABC . Suppose $L(A) = 1, L(B) = 2, L(C) = 3$. By the $n = 1$ statement, there must be an odd number of rainbow 1-faces on the edge AB .

Consider the cell graph of T with vertices V , where an edge is drawn between two vertices if and only if their corresponding cells share a rainbow 1-face. Note that $\deg(v_{out})$ is odd since AB has an odd number of rainbow 1-faces while BC and CA have none. Additionally, other vertices in the cell graph can have odd degree if and only if their corresponding cell is rainbow. This is true because a $1 - 2 - 3$ cell has degree 1, while $1 - 1 - 2$ and $1 - 2 - 2$ cells have degree 2, and the rest have degree 0.

Now, consider trails of the cell graph starting at v_{out} . Note that the path must end because the graph is finite, and it can only end at a vertex of odd degree, which is exactly the vertices of rainbow cells and v_{out} . Let there be A trails that end at v_{out} (which must include 2 edges incident to v_{out}) and B trails that end at a vertex of a rainbow cell (which must include only 1 edge incident to v_{out}). This gives

$$2A + B = \deg(v_{out}) = \text{odd number} \implies B \text{ is odd,}$$

so there exists an odd number of rainbow cells.

$n > 2$

Similarly, we construct the cell graph with vertices V such that there is an edge between two vertices if and only if their corresponding regions share a rainbow $(n - 1)$ -face. Note that the edges connected to v_{out} correspond to rainbow $(n - 1)$ -faces on the boundary of

Δ^n . For the rainbow $(n-1)$ -face colored using $[n]$, the simplicial subdivision of this face contains an odd number of rainbow $(n-1)$ -faces by the inductive hypothesis. The other $(n-1)$ -faces have no edges connected to v_{out} . Thus, $\deg(v_{out})$ is odd.

Now, we consider the degree of a vertex corresponding to an internal n -cell C . We do this by examining the $(n-1)$ -faces composed of n of the $n+1$ vertices on ∂C :

1. **Case 1:** $[n] \not\subseteq L(C)$. Then C cannot possibly have a rainbow $(n-1)$ -face. Thus, $\deg(C) = 0$.
2. **Case 2:** $[n] \subseteq L(C)$, but $n+1 \notin L(C)$. Because C has $n+1$ vertices, exactly one color from the set $[n]$ must be duplicated. Dropping either of the two vertices with the duplicated color yields a rainbow $(n-1)$ -face. Thus, $\deg(C) = 2$.
3. **Case 3:** $[n+1] = L(C)$. Then dropping the single vertex colored $n+1$ yields the unique rainbow $(n-1)$ -face. Thus, $\deg(C) = 1$.

By considering trails starting at v_{out} , it remains true that trails can only end at vertices of rainbow cells or v_{out} . If there are A trails of the former type, and B trails of the latter type, we still have

$$2A + B = \deg(v_{out}) = \text{odd number} \implies B \text{ is odd.}$$

Therefore, there must be an odd number of rainbow n -cells in the subdivision. This completes the induction. $\square$

3 Brouwer's Theorem

The classical proof of Brouwer's Fixed Point Theorem uses homology, or degree theory, to show that a continuous map from the simplex to itself without a fixed point would induce a continuous retraction from the simplex onto its boundary, leading to a contradiction [Bro11]. Instead of relying on algebraic topology, we present an equivalent combinatorial proof through Sperner's Lemma; a recent University of Chicago REU treatment develops the same general strategy [Gup25].

3.1 Proof of Brouwer's Theorem

Lemma 12 (Brouwer's Theorem on a Simplex). Let $\Delta^n \in \mathbb{R}^{n+1}$ be the n -dimensional simplex defined by the $n+1$ vertices

$$(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1).$$

Then any continuous function $f : \Delta^n \rightarrow \Delta^n$ has a fixed point. That is, $f(p) = p$.

Proof. For the sake of contradiction, suppose there is no fixed point. For any point $p \in \Delta$, write p in coordinate form:

$$\begin{aligned} p &= (p_1, \dots, p_{n+1}), \\ f(p) &= (f_1(p), \dots, f_{n+1}(p)), \\ g_i(p) &= f_i(p) - p_i. \end{aligned}$$

In particular, $p_i \geq 0$ and $\sum_{i=1}^{n+1} p_i = 1 \implies \sum_{i=1}^{n+1} g_i(p) = 0$. We choose to color the point p with color i that minimizes $g_i(p)$, with ties broken by choosing the smallest such index i . Since $\sum_{i=1}^{n+1} g_i(p) = 0$ and we assume there are no fixed points, the minimum must be strictly negative, so $g_i(p) < 0$. To show that this coloring is a valid Sperner coloring:

- Note that we must color the vertex $p = (0, \dots, 1, \dots, 0)$ that satisfies $p_i = 1$ with the color i , which satisfies the condition that each of the $n + 1$ vertices of Δ^n must be colored differently.
- Let p lie on a face F composed of vertices $v_{a_1} \dots v_{a_k}$, where $A = \{a_1, \dots, a_k\} \subset [n + 1]$. This implies that for all $a \notin A$,

$$p_a = 0 \implies f_a(p) \geq 0 = p_a \implies g_a(p) \geq 0.$$

Because the minimum $g_i(p)$ is strictly negative, the minimizing color i cannot be any $a \notin A$. Thus, $i \in A$. This satisfies the condition that a point on a face must share a color with one of the vertices of that face.

Thus, [Lemma 11](#) implies that there exists some rainbow cell in this simplicial subdivision.

Now, consider a sequence of simplicial subdivisions $S = \{S_1, S_2, \dots\}$ on Δ^n . We know there exists a rainbow cell in S_1 . Performing the simplicial subdivision S_2 on this rainbow cell, we again know there exists a smaller rainbow cell within it. Continue performing subsequent subdivisions on the resulting rainbow cell. Note that with each subdivision, the vertices of each cell get closer together, so the vertices of the rainbow cell grow arbitrarily close to each other as well.

Specifically, let $w_{i,j}$ denote the vertex of the rainbow cell of S_j with color i . Consider the sequence $\{w_{i,j}\}$ for fixed i and $j \rightarrow \infty$. Because Δ^n is compact, the sequence is bounded, and by the Bolzano-Weierstrass Theorem, this bounded sequence has a convergent subsequence. If $\{m_k\}$ is the set of indices of this convergent subsequence, let $\lim_{k \rightarrow \infty} w_{i,m_k} = w^*$. Because the vertices of the rainbow cell grow arbitrarily close to each other, we have that for all colors $i \in [n + 1]$,

$$\lim_{k \rightarrow \infty} w_{i,m_k} = w^*.$$

To conclude, because g_i is continuous and $g_i(w_{i,m_k}) \leq 0$, we have $g_i(w^*) \leq 0$ for all i . Thus, for all i ,

$$0 = \sum_{i=1}^{n+1} g_i(w^*) \leq 0 \implies g_i(w^*) = 0 \implies g(w^*) = 0.$$

Hence, $f(w^*) = w^*$, so w^* is a fixed point. This contradicts the original assumption and finishes the proof. $\square$

Theorem 13 (Brouwer's Fixed Point Theorem). Let K be a nonempty compact convex subset of $\mathbb{R}^n$. For any continuous function $f : K \rightarrow K$, there exists a fixed point $f(p) = p$.

Proof. We wish to show that there exists a continuous retraction from $\Delta \in \mathbb{R}^n$ to K , and then apply [Lemma 12](#). Consider enclosing the compact space K in a sufficiently large simplex Δ . Define $r : \Delta \rightarrow K$ as the retraction that takes $p \in \Delta$ to the closest point in K under the Euclidean norm. We wish to show that $r(p)$ is well-defined, unique, and continuous.

To show that r is well-defined, we note that K is nonempty and closed, and that the Euclidean norm is continuous. To show uniqueness, suppose $q_1 \neq q_2$ both have minimal distance from p and satisfy $\|p - q_1\| = \|p - q_2\|$. Then the midpoint $m = \frac{q_1 + q_2}{2} \in K$ by convexity, and

$$\|p - m\|^2 = \frac{1}{2} \|p - q_1\|^2 + \frac{1}{2} \|p - q_2\|^2 - \frac{1}{4} \|q_1 - q_2\|^2 < \|p - q_1\|^2,$$

contradicting the assumption that q_1, q_2 have minimal distance to p .

To show continuity, note that by the defining property of the closest-point retraction,

$$(p - r(p)) \cdot (r(q) - r(p)) \leq 0,$$

$$\begin{aligned}
(r(q) - q) \cdot (r(q) - r(p)) &\leq 0, \\
(p - q - (r(p) - r(q))) \cdot (r(q) - r(p)) &\leq 0, \\
\|r(p) - r(q)\|^2 &\leq (p - q) \cdot (r(p) - r(q)) \\
&\leq \|p - q\| \|r(p) - r(q)\|, \\
\|r(p) - r(q)\| &\leq \|p - q\|,
\end{aligned}$$

which shows that as q gets closer to p , $r(q)$ gets closer to $r(p)$, implying that r is continuous. To finish the proof, let $f : K \rightarrow K$ be a continuous function and define $g = f \circ r : \Delta \rightarrow \Delta$. By [Lemma 12](#), g has a fixed point p , but note that the codomain of f is K , so $p \in K$. Now by definition of retraction, $r(p) = p$, so

$$p = g(p) = f(r(p)) = f(p),$$

implying that p is a fixed point of f as well. □

3.2 The Other Direction

It is also possible to show the converse direction: Brouwer's Fixed Point Theorem implies a weak version of Sperner's Lemma.

Lemma 14 (Weak Sperner's Lemma). Every Sperner coloring of a simplicial subdivision of Δ contains a *rainbow* cell.

Proof. Let S be a simplicial subdivision of Δ such that there does not exist a rainbow cell, and let S^0 and Δ^0 be the sets of vertices of S and Δ , respectively. Define $f : S^0 \rightarrow \Delta^0$ as $f(v) = v_{L(v)+1}$. Then we extend this function to $F : \Delta \rightarrow \Delta$ by

$$F(p) = \sum_{i=1}^{n+1} p_i f(v_i).$$

F is well-defined and continuous. Note that if S has no rainbow cell, then $f(v_1), \dots, f(v_{n+1})$ are not all different. This implies that $F(p)$ has some coordinate equal to zero, and thus $\text{im}(F) = \partial\Delta$. Conversely, if $p \in \partial\Delta$, there exist faces $u_1 \dots u_k$ and $v_{i_1} \dots v_{i_m}$ for $1 \leq k < m < n + 1$ such that $p \in u_1 \dots u_k \subset v_{i_1} \dots v_{i_m}$. Then we can write $p = \sum_{j=1}^k p_j u_j$ and see that

$$F(p) = \sum_{j=1}^k p_j f(u_j) = \sum_{j=1}^k p_j v_{L(u_j)+1} \in v_{i_1+1} \dots v_{i_m+1}$$

because we established $L(u_j) \in \{i_1, \dots, i_m\}$ for all $j \in [k]$. Thus, F has no fixed point in $\partial\Delta$, but recall that $\text{im}(F) = \partial\Delta$, so this shows that F has no fixed point at all. But this contradicts Brouwer's Fixed Point Theorem because F must have a fixed point. □

Remark 15. In the full version of Sperner's Lemma, we get a result about the parity of rainbow cells, a type of result that is “incompatible” with the result about fixed points in Brouwer's Fixed Point Theorem, which is only about the existence of a fixed point. Thus, we can only prove this weaker version of Sperner's Lemma.

3.3 An Elementary Analytic Proof

Here we present a proof of Brouwer's Fixed Point Theorem using the Hairy Ball Theorem, following Milnor's analytic approach [[Mil78](#)].

Theorem 16 (Brouwer's Fixed Point Theorem). Every continuous map $\mathbf{f}$ from the unit ball in $\mathbb{R}^n$ $\mathbb{D}^n$ to itself possesses at least one fixed point. That is, there exists a point $\mathbf{x} \in \mathbb{D}^n$ such that $\mathbf{f}(\mathbf{x}) = \mathbf{x}$.

Proof. We proceed by contradiction. Assume such a mapping $\mathbf{f}$ exists, and consider $\mathbf{v}(\mathbf{x}) = \mathbf{x} - \mathbf{f}(\mathbf{x})$. The boundary of $\mathbb{D}^n$ is equivalent to $\mathbb{S}^{n-1}$, and can be defined as $\mathbb{S}^{n-1} = \{\mathbf{x} \mid \mathbf{x} \in \mathbb{D}^n, \|\mathbf{x}\| = 1\}$. For points $\mathbf{x} \in \mathbb{S}^{n-1}$, since $\|\mathbf{f}(\mathbf{x})\| \leq 1$, $\mathbf{v}(\mathbf{x})$ must be in the same n -hemisphere as $\mathbf{x}$, and thus

$$\mathbf{x} \cdot \mathbf{v}(\mathbf{x}) > 0 \quad \forall \mathbf{x} \in \mathbb{S}^{n-1}.$$

Now, consider

$$\mathbf{w}(\mathbf{x}) = \mathbf{x} - \mathbf{f}(\mathbf{x}) \cdot \frac{1 - \mathbf{x} \cdot \mathbf{x}}{1 - \mathbf{x} \cdot \mathbf{f}(\mathbf{x})}.$$

If $\|\mathbf{x}\| = 1$, $\mathbf{w}(\mathbf{x}) = \mathbf{x} \neq 0$. Otherwise, if $\mathbf{x}$ and $\mathbf{f}(\mathbf{x})$ are linearly independent, then from the above $\mathbf{w}(\mathbf{x}) = \mathbf{x} - c\mathbf{f}(\mathbf{x}) \neq 0$ where c is a scalar. Finally, if $\mathbf{x}$ and $\mathbf{f}(\mathbf{x})$ are linearly dependent, then

$$\mathbf{w}(\mathbf{x}) = \frac{\mathbf{x}(1 - \mathbf{x} \cdot \mathbf{f}(\mathbf{x})) - \mathbf{f}(\mathbf{x})(1 - \mathbf{x} \cdot \mathbf{x})}{1 - \mathbf{x} \cdot \mathbf{f}(\mathbf{x})} = \frac{\mathbf{x} - \mathbf{f}(\mathbf{x})}{1 - \mathbf{x} \cdot \mathbf{f}(\mathbf{x})} \neq 0.$$

Hence, $\mathbf{w}(\mathbf{x})$ is a nonzero continuous vector field on $\mathbb{D}^n$ which points directly outward on the boundary.

Next, we consider the stereographic projection $\mathbf{s}(\mathbf{x})$ of $\mathbf{w}(\mathbf{x})$ from $\mathbb{D}^n$ to $\mathbb{S}^n \subset \mathbb{R}^{n+1}$. First, we project from the north pole of $\mathbb{S}^n$ $(0, 0, \dots, 0, 1)$ to the southern hemisphere of $\mathbb{S}^n$, that is, all points $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n+1})$ where $\mathbf{x}_{n+1} \leq 0$. The projection formula is

$$\mathbf{s}(\mathbf{x}) = \left(\frac{2\mathbf{x}_1}{\mathbf{x} \cdot \mathbf{x} + 1}, \frac{2\mathbf{x}_2}{\mathbf{x} \cdot \mathbf{x} + 1}, \dots, \frac{2\mathbf{x}_n}{\mathbf{x} \cdot \mathbf{x} + 1}, \frac{\mathbf{x} \cdot \mathbf{x} - 1}{\mathbf{x} \cdot \mathbf{x} + 1} \right).$$

Let $\mathbf{s}(\mathbf{x}) = \mathbf{u}$. To project the vector field $\mathbf{w}(\mathbf{x})$ onto $\mathbb{S}^n$ (as a tangent field), we have

$$\mathbf{W}(\mathbf{u}) = \frac{d}{dt} \mathbf{s}(\mathbf{x} + t\mathbf{w}(\mathbf{x}))$$

at $t \rightarrow 0$. Consider the points on $\mathbb{S}^{n-1}$ (i.e. on the equator of $\mathbb{S}^n$), which have the property that $\mathbf{w}(\mathbf{x}) = \mathbf{x}$. Then,

$$\begin{aligned} \mathbf{W}(\mathbf{u}) &= \frac{d}{dt} \mathbf{s}((1+t)\mathbf{x}) = \frac{d}{dt} \left(\frac{2(1+t)\mathbf{x}_1}{(1+t)^2 + 1}, \frac{2(1+t)\mathbf{x}_2}{(1+t)^2 + 1}, \dots, \frac{2(1+t)\mathbf{x}_n}{(1+t)^2 + 1}, \frac{(1+t)^2 - 1}{(1+t)^2 + 1} \right) \\ &= (0, 0, \dots, 0, 1). \end{aligned}$$

Now, consider a stereographic projection from the south pole $(0, 0, \dots, 0, -1)$ to the northern hemisphere of $\mathbb{S}^n$, but projecting $-\mathbf{w}(\mathbf{x})$ to a tangent field (say $\mathbf{W}'(\mathbf{x})$ instead of $\mathbf{w}(\mathbf{x})$). Once again, we can find that for points on $\mathbb{S}^{n-1}$, $\mathbf{W}'(\mathbf{x}) = (0, 0, \dots, 0, 1)$. Hence, we can compose these two vector fields to create a non-zero continuous tangent vector field $\mathbf{V}$ everywhere on $\mathbb{S}^n$.

If n is even, this is impossible by the Hairy Ball Theorem. If n is odd and equal to $2k - 1$ for some integer k , assume a map $\mathbf{f}$ exists mapping $\mathbb{D}^{2k-1}$ to itself without any fixed points. This would then produce a mapping

$$\mathbf{F}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{2k}) = (\mathbf{f}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{2k-1}), 0)$$

without any fixed points, a contradiction. □

4 Borsuk–Ulam Theorem and Equivalences

4.1 Borsuk–Ulam Theorem

Theorem 17 (Borsuk–Ulam Theorem [Bor33]). For any continuous function $f : S^n \rightarrow \mathbb{R}^n$, there exists a pair of antipodal points $\{p, -p\} \in S^n$ such that $f(p) = f(-p)$.

Like Brouwer’s theorem, the standard proof of the Borsuk–Ulam Theorem uses algebraic topology, typically through homology or degree theory. A combinatorial alternative is provided by Ky Fan’s Lemma [Fan52], which concerns antipodal labelings of triangulations of the sphere and imposes conditions on complementary edges. By constructing an appropriate antipodal labeling from a continuous map, one can use Ky Fan’s Lemma to derive the Borsuk–Ulam Theorem (first reducing to an earlier discovered corollary, Tucker’s Lemma).

4.2 Equivalences

While we have shown that there are combinatorial methods for proving Brouwer’s Fixed Point Theorem, we now remark on a general set of equivalences between fixed-point theorems and combinatorial theorems. Specifically, Brouwer’s Fixed Point Theorem is the topological equivalent of Sperner’s Lemma, and the Borsuk–Ulam Theorem is the topological equivalent of Ky Fan’s Lemma.

$$\begin{array}{ccc} \text{Borsuk–Ulam} & \implies & \text{Brouwer} \\ \Downarrow & & \Downarrow \\ \text{Ky Fan} & \implies & \text{Sperner} \end{array}$$

In particular, we have that Borsuk–Ulam/Ky Fan is a stronger version of Brouwer/Sperner. We will show the topological portion of this implication below, following the direct construction of Su [Su97].

Claim 18. The Borsuk–Ulam Theorem implies Brouwer’s Fixed Point Theorem.

Proof. To prove the existence of a fixed point for the continuous function $f : B^n \rightarrow B^n$, we seek to construct a map $g : S^n \rightarrow \mathbb{R}^n$ that is antipode-preserving, that is, $g(p) = -g(-p)$ for all $p \in S^n$. If we can find a point p with $g(p) = 0$, this implies that f has a fixed point. Then we can retract the ball B^n onto Δ in a similar manner, and the rest of the proof proceeds as before.

Let $g(p) = (p_1, \dots, p_n)$ and $g_i(p) = p_i$. First, note that S^n is homeomorphic to the boundary of the $(n + 1)$ -dimensional cube

$$B^n = \{(p_1, \dots, p_{n+1}) \mid |p_i| \leq 1 \text{ and } \max |p_i| = 1\}$$

by considering rays from the origin. Furthermore, define

$$\begin{aligned} B_1^n &= \{p \mid p = (p_1, \dots, p_n, 1)\}, \\ B_0^n &= \{p \mid p = (p_1, \dots, p_n, 0)\}, \\ B_{-1}^n &= \{p \mid p = (p_1, \dots, p_n, -1)\} \end{aligned}$$

as the “top”, “equator”, and “bottom” planes of the $(n + 1)$ -dimensional cube. Also, define $h(p) = (p_1, \dots, p_n)$, so that h ignores the last coordinate of p .

We will now define g such that g is continuous and $g(p) = -g(-p)$ for all $p \in B^n$. Specifically,

$$g(p) = \begin{cases} h(p) - f(h(p)) & p_{n+1} = 1 \\ h(p) + f(-h(p)) & p_{n+1} = -1 \\ h(p) + \frac{g(p_1, \dots, p_n, 1) + g(p_1, \dots, p_n, -1)}{2} & p_{n+1} = 0 \\ p_{n+1} \cdot g(p_1, \dots, p_n, 1) + (1 - p_{n+1}) \cdot g(p_1, \dots, p_n, 0) & p_{n+1} \in [0, 1] \\ -p_{n+1} \cdot g(p_1, \dots, p_n, -1) + (1 + p_{n+1}) \cdot g(p_1, \dots, p_n, 0) & p_{n+1} \in [-1, 0] \end{cases}$$

After defining $g(p)$ on the “top” and “bottom” planes, the equator is defined by averaging the value of g on the top and bottom planes, and then lifting it by the value $h(p)$. Then the rest of B^n is defined by linearly interpolating the values from B_0^n to B_1^n and B_{-1}^n .

We wish to show that for any p such that $|p_{n+1}| < 1$, $g(p) \neq 0$. Note that in this case, there exists some k such that $|p_k| = 1$. One can check that $g_k(p_1, \dots, p_n, \pm 1)$ and $g_k(p_1, \dots, p_n, 0)$ all have the same sign as p_k , and the latter is also nonzero. Also, $(1 - p_{n+1}), (1 + p_{n+1})$ are positive because $|p_{n+1}| < 1$, so the definition of g implies that $g(p)$ is nonzero in this case.

Note that Borsuk–Ulam states that there exists some p such that $g(p) = g(-p) \implies g(p) = 0$. But by the work above, we have shown that $|p_{n+1}| = 1$, so either $q = p$ or $q = -p$ satisfies $q \in B_1^n$. Then $0 = g(q) = h(q) - f(h(q))$ implies that $h(q) \in B^n$ is a fixed point of f .

□

5 Applications of Fixed Point Theorems

5.1 Necklace Splitting Problem

This problem involves a necklace string attached with n types of jewels, which must be split evenly between 2 thieves, A and B . The number of jewels of the j -th type, a_j , is guaranteed to be a multiple of n . Firstly, we will consider a continuous version of this problem, where individual jewels can be divided, before generalizing to the discrete version.

Consider the necklace as a string of jewels with length 1, and take a point $(x_0, x_1, x_2, \dots, x_n) \in \mathbb{S}^n$. Since $x_0^2 + x_1^2 + \dots + x_n^2 = 1$, we can take n cuts of length $x_0^2, x_1^2, x_2^2, \dots$. To determine which thief to give the i -th piece, we use the sign of x_i .

Now, consider a fairness function $f : \mathbb{S}^n \rightarrow \mathbb{R}$, where $f(x)_j$ represents the difference in the amount of the j -th jewel type given to thief A and thief B . The Borsuk–Ulam theorem states that there must exist a point $x \in \mathbb{S}^n$ such that $f(x) = f(-x)$, meaning that if the two thieves swapped allocations, the inequality would be the same. This implies that they had exactly the same amount of each jewel type, using at most n cuts.

Remark: This can be extended to 2^k thieves. Suppose we can solve the problem for 2^{k-1} thieves. Then, we can first divide the necklace into 2 equal groups of jewels, then divide each group into 2^{k-1} equal allocations for each thief.

Remark: Using the more powerful Bárány–Shlosman–Szűcs Theorem, we can solve for any prime number of thieves (say p) using $(p-1)n$ cuts. A similar logic can be applied as in the previous remark, where a a -thief solution and a b -thief solution can be combined into a ab -thief solution. With m thieves, at most $(m-1)n$ cuts are required.

To convert this into a discrete problem, where the number of jewels of each type is a multiple of the number of thieves m and each thief may only take whole jewels, we must remove all *bad cuts* which land in the interior of a jewel. We can create a multigraph G with vertices $v_1, v_2, \dots, v_m$ representing each thief. For every bad cut through a jewel of type i , we can draw an edge of color i between the two thieves whose assigned jewels meet at that cut.

Consider each subgraph G_i of G containing only edges of color i . Observe that each vertex with positive degree has degree at least 2, otherwise a thief would not have a whole number of the i -th type of jewel. Thus, the graph must contain a cycle. Roughly speaking, we may now ‘shift’ all cuts corresponding to edges of the cycle by the same amount until we reach a jewel boundary. As this keeps the allocations constant (the gain from shifting one cut is equal to the loss of shifting another cut), we can use this to repeatedly remove bad cuts without changing the allocation amounts. This will result in a valid discrete split.

5.2 Ham Sandwich Theorem

We state the general case of $n \geq 2$ dimensions, and show the case of only $n = 3$.

Theorem 19 (Ham Sandwich). Any n bounded, compact sets in $\mathbb{R}^n$ can be simultaneously bisected by an $(n - 1)$ -dimensional hyperplane.

Proof for $n = 3$. The theorem can be rephrased in three dimensions as follows. Let A , B , and C be compact subsets of $\mathbb{R}^3$. There exists a plane that simultaneously divides each of A , B , and C into two sets of equal volume.

Let x be a unit vector in $\mathbb{R}^3$, and let P_x be a plane with normal vector x , which divides A into 2 sets of equal volume. Then, denote P_x^+ and P_x^- to be the half space above and below P_x respectively (i.e. that x points into and away from). Finally, define

$$f(x) = (\text{Vol}(P_x^+ \cap B), \text{Vol}(P_x^+ \cap C)).$$

It can be shown that f is continuous. By the Borsuk–Ulam theorem, there then exists an x such that $f(x) = f(-x)$, meaning that

$$(\text{Vol}(P_x^+ \cap B), \text{Vol}(P_x^+ \cap C)) = (\text{Vol}(P_{-x}^+ \cap B), \text{Vol}(P_{-x}^+ \cap C))$$

$$\text{Vol}(P_x^+ \cap B) = \text{Vol}(P_x^- \cap B), \quad \text{Vol}(P_x^+ \cap C) = \text{Vol}(P_x^- \cap C).$$

□

Remark: The Ham Sandwich Theorem generalizes naturally to any dimension $n \geq 2$, using the Borsuk–Ulam theorem.

5.3 Existence of Nash Equilibria

We will prove the existence of a Nash equilibrium in multiplayer games as described in [Kar17], where players play independently and without knowledge of other players’ strategies.

Definition 20. A *pure strategy* is an option that a player may play in a game. For example, in rock paper scissors, the pure strategies are rock, paper, and scissors.

Definition 21. A *utility function* is a map from a given play in a game to the set of all combinations of strategies of players to the utility of one particular player. For example, in rock paper scissors, the utility function maps combinations of plays to victories, draws, and losses.

Definition 22. An n -player *game* $G := (A, u)$ is a set of player strategies and utility functions. It has players $\{1, 2, \dots, n\}$, the i -th of which has a set of pure strategies A_i and a utility function $u_i : A \rightarrow \mathbb{R}$. The set of pure strategies for the entire game is $A := \prod_{i=1}^n A_i$ and the overall utility function is defined as $u := \prod_{i=1}^n u_i$.

Definition 23. A *mixed strategy* for a player i is the strategy of randomly playing one of their pure strategies with some probability. Specifically, if $A_i = \{a_1, a_2, \dots, a_k\}$ and $\langle \lambda_1, \lambda_2, \dots, \lambda_k \rangle \in [0, 1]^k$ with $\sum_{i=1}^k \lambda_i = 1$, then define $M_i = \langle \lambda_1, \lambda_2, \dots, \lambda_k \rangle$ to be their mixed strategy set and $M = \prod_{i=1}^n M_i$ to be the set of all combinations of each player's mixed strategies.

Definition 24. We can extend the definition of u_i to map $M \rightarrow \mathbb{R}$, where u_i of a mixed strategy is the sum of utilities of each possible set of pure strategies multiplied by the probability that each combination will be played.

Definition 25. Given a strategy set $\vec{a} = \langle a_1, \dots, a_n \rangle \in M$, write $\vec{a}_{-i} = \langle a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n \rangle$.

Definition 26. A *Nash equilibrium* is a set of strategies $\vec{a}^* = \langle a_1^*, a_2^*, \dots, a_n^* \rangle \in M$ such that for $i \in \{1, 2, \dots, n\}$ and all $\hat{a}_i \in A_i$

$$u_i(\vec{a}^*) \geq u_i(\vec{a}_{-i}^*, \hat{a}_i).$$

Less formally, this is a set of strategies with the property that a player changing strategy would not increase their utility.

Definition 27. The *best response* of a given strategy is the utility-maximizing strategy for player i given the strategies of all other players $\vec{a} \in M$, and is written as

$$BR_i(\vec{a}_{-i}) := \{a \in M_i \mid u_i(\vec{a}_{-i}, a) = \max_{\hat{a} \in M_i} u_i(\vec{a}_{-i}, \hat{a})\}.$$

Definition 28. Lastly, define $BR : M \rightarrow M$ with $\vec{a} \in M$ as

$$BR(\vec{a}) = \prod_{i=1}^n BR_i(\vec{a}_{-i}).$$

By the previous definitions, if $\vec{a} \in BR(\vec{a})$, then $\vec{a}$ is a Nash equilibrium.

The following proof of the Nash equilibrium will involve constructing a function taking an arbitrary set of mixed strategies, and returning a set of mixed strategies with equal or better utility. We will then use Brouwer's Fixed Point Theorem to show that we will reach a set of mixed strategies where no player can improve their utility, hence showing the existence of a Nash equilibrium. We will also assume that each player has a finite strategy set A_i .

Lemma 29. For every strategy $\vec{a} \in M$ and all $i \in \{1, 2, \dots, n\}$, there exists some pure strategy in $BR_i(\vec{a}_{-i})$.

Proof. Let $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle \in M$, and consider $u_i(a_i, \vec{a}_{-i})$. Since there are a finite number of pure strategies in A_i , there exists a pure strategy that maximizes player i 's utility. That is, if $e_j \in M_i$ has j -th coordinate 1 and all other coordinates 0, there exists a unit vector $e^* \in M_i$ such that

$$u_i(e^*, \vec{a}_{-i}) = \max_{j \in \{1, 2, \dots, |A_i|\}} u_i(e_j, \vec{a}_{-i}).$$

Now consider any mixed strategy $a' = \langle \lambda_1, \lambda_2, \dots, \lambda_{|A_i|} \rangle \in M_i$. The utility function of a mixed strategy is simply a linear combination of the utilities of pure strategies, which we can find:

$$u_i(a', \vec{a}_{-i}) = \sum_{j=1}^{|A_i|} \lambda_j u_i(e_j, \vec{a}_{-i}) \leq \sum_{j=1}^{|A_i|} \lambda_j u_i(e^*, \vec{a}_{-i}) = u_i(e^*, \vec{a}_{-i}).$$

Thus, $e^* \in BR_i(\vec{a}_{-i})$. □

We now construct a continuous function on mixed strategies which moves them towards the Nash equilibrium.

Construction: Let $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle \in M$ be a set of mixed strategies, and let $a_i = \langle \alpha_{i,1}, \alpha_{i,2}, \dots, \alpha_{i,|A_i|} \rangle$ be player i 's mixed strategy. Now, define $\phi_{ij} : M \rightarrow \mathbb{R}$ to be how much higher utility the i -th player can obtain by instead playing with the j -th pure strategy:

$$\phi_{ij}(\vec{a}) = \max\{0, u_i(e_j, \vec{a}_{-i}) - u_i(\vec{a})\}.$$

Next, define $b_i : M \rightarrow M_i$ to give the movement to a better strategy of player i , given by

$$b_i(\vec{a}) = \left\langle \frac{\alpha_{i,1} + \phi_{i1}(\vec{a})}{1 + \sum_{j=1}^{|A_i|} \phi_{ij}(\vec{a})}, \dots, \frac{\alpha_{i,|A_i|} + \phi_{i|A_i|}(\vec{a})}{1 + \sum_{j=1}^{|A_i|} \phi_{ij}(\vec{a})} \right\rangle.$$

In particular, if implementing a fixed strategy e_j yields a higher proportion of utility ϕ_{ij} (out of all fixed strategies $\sum \phi_{ij}$) compared to the frequency in which that strategy appears in the current mixed strategy, then the function raises the probability of using that strategy, and vice versa.

Finally, define $b : M \rightarrow M$ to be the composition of all better strategies

$$b(\vec{a}) = \langle b_1(\vec{a}), b_2(\vec{a}), \dots, b_n(\vec{a}) \rangle.$$

Lemma 30. b is continuous and well-defined.

Proof. To show that b is well-defined, observe that

$$\sum_{k=1}^{|A_i|} \frac{\alpha_{i,k} + \phi_{ik}(\vec{a})}{1 + \sum_{j=1}^{|A_i|} \phi_{ij}(\vec{a})} = \frac{\sum_{k=1}^{|A_i|} \alpha_{i,k} + \sum_{k=1}^{|A_i|} \phi_{ik}(\vec{a})}{1 + \sum_{j=1}^{|A_i|} \phi_{ij}(\vec{a})} = \frac{1 + \sum_{k=1}^{|A_i|} \phi_{ik}(\vec{a})}{1 + \sum_{j=1}^{|A_i|} \phi_{ij}(\vec{a})} = 1.$$

This implies that each element of $b_i(\vec{a})$ is less than or equal to 1. Moreover, since $a_{ij} \in [0, 1]$ and $\phi_{ij} \geq 0$, we have

$$0 \leq \frac{\alpha_{i,k} + \phi_{ik}(\vec{a})}{1 + \sum_{j=1}^{|A_i|} \phi_{ij}(\vec{a})} \leq 1.$$

Hence, the probabilities for the new mixed strategy are valid (all between 0 and 1 while summing to 1), so we can conclude that $b_i(\vec{a}) \in M_i$, and so $b(\vec{a}) \in M$, meaning it is well-defined.

Next, to show that b is a continuous map, observe that u_i is a linear combination of u_i for a set of pure strategies, so it is also continuous with respect to small changes in probability. Hence, ϕ_{ij} is continuous as it is defined as the difference of continuous functions. Lastly, each element of b 's output is a linear combination of ϕ_{ij} , so b is continuous. $\square$

Lemma 31. For all strategies $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle \in M$, if $b(\vec{a}) = \vec{a}$ then $b(\vec{a}) \in BR(\vec{a})$.

Proof. If $b(\vec{a}) = \vec{a}$, then for all $i \in \{1, 2, \dots, n\}$ and $j \in \{1, 2, \dots, |A_i|\}$, we have $\phi_{ij}(\vec{a}) = 0$. Hence, no pure strategy gives higher utility than the current mixed strategy $\vec{a}$. However, by Lemma 29, there is one pure strategy that maximizes utility, and so we know that for each player i , a_i is already their best response with regard to other players using their respective strategies in $\vec{a}$. Hence

$$\vec{a} = b(\vec{a}) \in BR(\vec{a}). \quad \square$$

$\square$

Theorem 32 (Nash’s Theorem). Every n -player game with a finite number of pure strategies has a Nash equilibrium.

Because from Lemma 30 we have that $b : M \rightarrow M$ is continuous, using Brouwer’s Fixed Point Theorem, b has a fixed point, which means that in all games, there is some set of mixed strategies $\vec{a} \in M$ such that $\vec{a} = b(\vec{a}) \in BR(\vec{a})$, which is a Nash equilibrium. $\square$

Remark: This proof only applies to games where each player has a finite strategy set. For infinite strategy sets, more advanced theorems such as Kakutani’s Fixed Point Theorem and Glicksburg’s Fixed Point Theorem, which deal with infinite-dimensional set-valued correspondences.

5.4 Cake and Rent Division

Fair division problems are those involving splitting assets among multiple parties such that each of them considers the outcome to be fair and desirable [Su99]. The easier version of this problem is cake division, in which n people wish to divide a rectangular cake with $n - 1$ parallel cuts perpendicular to its width. Assuming the width of the cake to be 1, and the size of the i -th piece by x_i , then the space defined by $x_1 + x_2 + \cdots + x_n = 1$ and $x_i \geq 0$ forms a $(n - 1)$ -simplex in $\mathbb{R}^n$.

We may use barycentric subdivision m times to create a triangulation of our simplex. Loosely speaking, each subdivision involves finding the barycenters in each dimension, and then connecting all pairs of barycenters of different dimensions such that one of their corresponding faces is fully contained inside the other.

This has the property of allowing us to assign one of the n people to each vertex such that each $(n - 1)$ -cell has n distinct vertices: label all vertices from the $(m - 1)$ -th subdivision the same, and assign a distinct person to each set of barycenters from the final subdivision that are of the same dimension.

We now provide two assumptions for each person’s preferences:

- Each person will always prefer a piece with mass to an empty piece.
- If a person prefers the same piece in a convergent sequence of cuts, then they will prefer that piece at the limiting cut.

Now, at each vertex of the subdivision, we may consider the set of cuts determined by that vertex, and ask the person assigned to that vertex: “Which slice of the cake would you prefer?”. By the first preference assumption, it is easy to see that the answers will form a valid Sperner coloring. Hence, there exists a cell in the subdivision where the answers at the vertices are all distinct. By taking finer and finer subdivisions, as the people around each cell are all distinct, we will find a convergent sequence of cuts such that at the limit, each person will prefer a different piece.

The cake-cutting method can be extended to address a discretized variation, the rent partitioning problem. In this problem, n housemates are renting out an n -bedroom house for some fixed total rent, with the following assumptions:

- In any partition of the rent, each person finds some room acceptable.
- Each person always prefers a free room over a non-free room.
- If a person prefers the same room in a convergent sequence of prices, then they will prefer that room at the limiting price.

If we construct a barycentric subdivision of our the $(n - 1)$ -simplex, assign a person to each vertex, and ask each person for their room preference given the rent cost of each room, then unlike before, it is not guaranteed to give a valid Sperner coloring (as people will always

prefer free rooms whereas they would always avoid nonexistent cake slices, so the situation is reversed).

However, consider the dualization of our pricing simplex, where k -dimensional faces map to $(n - 1 - k)$ -dimensional faces. This is in 1-1 correspondence with our original simplex, and can be barycentrically subdivided in exactly the same way. Moreover, this is now a valid Sperner coloring: each boundary of a dual face corresponds to an original corner vertex where all but one room was free, ensuring that room being avoided on that dual face. Hence, we can once again apply Sperner’s lemma to find a convergent sequence of rent allocations such that each person will prefer a different room.

6 Computing Nash Equilibria

6.1 Complexity Analysis

In this section, we will analyze what category of problem finding a Mixed Nash Equilibrium (MNE) for a 2-player game lies in. It turns out that this is an FNP (“functional NP”) problem, which is a class of problems where particular solutions cannot necessarily be found in polynomial time, but can be verified in polynomial time [Rou13]. This is because verifying a pair mixed strategies only requires checking if a player can profitably improve their strategy. As seen above, this only requires checking deviations to pure strategies, meaning that there are only a finite number of calculations which can be calculated in polynomial time.

More specifically, finding the MNE is part of the class TFNP (“total FNP”), where ‘total’ signifies that each problem has a guaranteed solution, which we proved above. Interestingly, the guarantee of a solution prevents the MNE problem from being FNP-complete (i.e. every FNP problem can be reduced to it) under standard assumptions. If it were, then we could reduce the problem of the unsatisfiability of a Boolean formula (UNSAT), a coNP problem to finding a Nash Equilibrium (coNP meaning the cases where the answer is ‘no’ are NP). However, complexity theorists strongly believe that $NP \neq coNP$, so we may conclude that MNE is not FNP-complete.

However, we cannot conclude that MNE is TFNP-complete either, as TFNP is a semantic complexity class, which is one not based off a fixed computational model but a property of the problems themselves (namely that they must have a solution for any reason at all). In fact, TFNP is believed not to contain any complete problems. Still, subclasses of TFNP have been established containing complete problems.

MNE lies in the subclass known as PPAD (“Polynomial Parity Argument on Directed graphs”), which we define using the following PPAD-complete problem, called **END OF THE LINE**:

An exponentially large directed graph G is defined by two circuits P and N taking in a node and returning another node, where P and N are the “possible previous” and “possible next” circuits respectively. They define G in that there is an edge from v_1 to v_2 if and only if $P(v_2) = v_1$ and $N(v_1) = v_2$ (which implies each node has both indegree and outdegree at most 1).

The **END OF THE LINE** problem is as follows: given a vertex v with no predecessor, find another vertex with either no predecessor or no successor.

It is easy to see that PPAD is a subset of FNP, as it is quick to verify whether a node has indegree different from its outdegree.

Claim: Finding a rainbow cell in a Sperner coloring is a PPAD problem.

Proof: It is sufficient to construct circuits P and N where a solution to **END OF THE LINE** maps to a solution to the Sperner problem. Recall that in the directed-walk proof of Sperner's lemma, we constructed a directed graph where each edge had indegree and outdegree at most 1. Moreover, there was one node that had no ingoing edge, which we can call the starting simplex, labeled s . We can informally define P and N with the following:

- Set $P(s) = s$, and $N(s)$ to be the simplex sharing a rainbow $(n - 1)$ -face with s .
- If a simplex t is not rainbow and also does not contain a rainbow $(n - 1)$ -face, set $P(t) = t$ and $N(t) = s$. This guarantees that t is isolated.
- If a simplex t has a rainbow $(n - 1)$ -face shared with another simplex t' , then either $P(t) = t'$ and $N(t') = t$, or $P(t') = t$ and $N(t) = t'$, i.e. there is a directed edge between t and t' . (Note: to determine the specific direction of the edge, a concept of *sign* of a face must be defined, see [DBP10])

It is clear that the graph created for **END OF THE LINE** is isomorphic to the graph constructed to prove Sperner's lemma. Hence the solutions to **END OF THE LINE** are precisely the solutions to Sperner's lemma, meaning we have successfully completed the reduction. $\square$

6.2 Scarf's Algorithm

Scarf's Algorithm can find what is known as a *weak approximation* of a fixed point for any continuous mapping f , that is, it finds a point x such that $\max_i |x_i - f(x)_i| < \epsilon$. (*Remark:* This is opposed to a *strong approximation*, which finds a point x such that there exists x^* such that $\|x - x^*\| < \epsilon$ and $x^* = f(x^*)$.)

First, we create a “restricted” subdivision of our simplex with k new vertices not on its boundary. Denote the vertices of the simplex to be $v_1, v_2, \dots, v_n$ and further vertices by $v_{n+1}, v_{n+2}, \dots, v_{n+k}$. Next, for each vertex v , we can associate a label $l(v)$ to each vertex using the following:

$$l(v_j) = \min_{1 \leq i < n} \{i \mid f(v_j)_i \leq v_{j_i}\}$$

where x_i is the i -th coordinate of x . Since $\sum x_i = 1$ for any point in our simplex, such a label must exist (otherwise $\sum f(x)_i > 1$). Note that for $j \in \{1, \dots, n\}$, we have $v_{j_i} = 1$ for $i = j$ and $v_{j_i} = 0$ otherwise, hence it must be that $l(v_j) = j$. Thus, l is a Sperner labeling. We now describe the procedure of the algorithm.

1. Begin from the unique cell containing vertices $v_2, v_3, \dots, v_n$, and an additional vertex v_{n+j} . Call this the *initial* cell.
2. If $l(v_{n+j}) = 1$, end the algorithm as we have found a rainbow cell, whose vertices are approximations of the fixed point.
3. Otherwise, eliminate the vertex sharing the same label as v_{n+j} . Since each $(n - 1)$ -face is shared by exactly 2 cells, we can now traverse to a new cell, say with a new vertex v_{n+k} . Repeat step 2.

Remark: Notice how if the algorithm terminates (in an extreme case) on the first step, then the resulting rainbow cell will have a very ‘wide’ face which is part of the original simplex. Scarf solved this by embedding the simplex in a higher dimensional simplex in such a way to form a restricted subdivision of that simplex while keeping a high level of refinement throughout the original simplex.

The error of this algorithm can be bounded. Let maximum diameter of our triangles is at most δ , the modulus of continuity of f is ϵ . If the fixed point is x^* and we have a vertex x of the rainbow simplex with label i , then

$$f(x^*)_i - x_i^* = (f(x^*)_i - f(x)_i) + (f(x)_i - x_i) + (x_i - x_i^*)$$

$$\leq \epsilon + 0 + \delta = \epsilon + \delta.$$

Hence, if we consider the lower bound of the error for any one coordinate, we see that it is bounded by $|f(x^*)_i - x_i^*| < n(\epsilon + \delta)$.

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